

HW 4 Math 638 Differential Geometry II, Winter 2026

Due on Wednesday, February 04, 2026 on class.

Q4.1. ([Lee, 2nd, 10-7, p.269]) Compute the transition function for TS^2 associated with the two local trivializations determined by stereographic coordinates in [Lee, 2nd, Problem 1-7].

Q4.2 ([Lee, 2nd, 10-3, p.269]) Let $\mathbf{VB}$ denote the category whose objects are smooth vector bundles and whose morphisms are smooth bundle homomorphism, and let $\mathbf{Diff}$ denote the category whose objects are smooth manifolds and whose morphisms are smooth maps. Show that the assignment $M \rightarrow TM, F \rightarrow dF$ defines a covariant functor from $\mathbf{Diff}$ to $\mathbf{VB}$, called the tangent functor.

(Hint: Let $\mathbf{C}$ and $\mathbf{D}$ be categories. A covariant functor F from $\mathbf{C}$ to $\mathbf{D}$ is a mapping that:

- associates each object X in $\mathbf{C}$ to an object $F(X)$ in $\mathbf{D}$,
- associates each morphism $f : X \rightarrow Y$ in $\mathbf{C}$ to a morphism $F(f) : F(X) \rightarrow F(Y)$ in $\mathbf{D}$ such that the following two conditions hold:

$$F(\text{id}_X) = \text{id}_{F(X)} \quad \text{for every object } X \text{ in } \mathbf{C},$$

$$F(g \circ f) = F(g) \circ F(f) \quad \text{for all morphisms } f : X \rightarrow Y \text{ and } g : Y \rightarrow Z \text{ in } \mathbf{C}.$$

From Wikipedia)

Q4.3 ([Lee, 2nd, 10-5 part, p.269]) Let $\pi : E \rightarrow M$ be a smooth vector bundle of rank k over a smooth manifold M^n without boundary. Suppose that $\{U_\alpha\}_{\alpha \in A}$ is an open cover of M , and for each $\alpha \in A$ we are given a smooth local trivialization $\Phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^k$ of E . For each $\alpha, \beta \in A$ such that $U_\alpha \cap U_\beta \neq \emptyset$, let $\tau_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow GL(k; \mathbb{R})$ be the transition function defined by [Lee, 2nd, (10.3)]. Show that the following identity is satisfied for all $\alpha, \beta, \gamma \in A$ with $U_\alpha \cap U_\beta \cap U_\gamma \neq \emptyset$

$$\tau_{\alpha\beta} \cdot \tau_{\beta\gamma} = \tau_{\alpha\gamma} \quad \text{for } p \in U_\alpha \cap U_\beta \cap U_\gamma.$$

Q4.4 ([Lee, 2ed, 7-11 related, p.560]) Note that the Hopf action of S^1 on S^{2n+1} is induced from $\mathbb{C}^*$ -action on $(\mathbb{C}^{n+1})^*$. Let $\pi : S^{2n+1} \rightarrow \mathbb{CP}^n$ be the projection induced from $(\mathbb{C}^{n+1})^* \rightarrow \mathbb{CP}^n$. Let $U = \{[1, z^1, \dots, z^n] \in \mathbb{CP}^n, (z^1, \dots, z^n) \in \mathbb{C}^n\}$. Construct a diffeomorphism $\Phi : \pi^{-1}(U) \rightarrow U \times S^1$ such that $\pi = \pi_1 \circ \Phi$ on $\pi^{-1}(U)$, where $\pi_1 : U \times S^1 \rightarrow U$ is the projection map.